# Uncertainty-Calibrated Virtual Training for Out-of-Distribution Meta-Reinforcement Learning

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**Abstract**

Virtual-task methods for meta-reinforcement learning expand the training distribution by synthesizing transitions for tasks the agent has not directly experienced, but the synthesized rewards and next states can be wrong under extrapolation, and training uniformly on such samples can bias Bellman targets and destabilize the critic. We propose Uncertainty-Calibrated Virtual Training (UC-VT), a local extension of Task-Aware Virtual Training (TAVT) that uses an independently initialized ensemble of task decoders to estimate epistemic uncertainty and down-weights unreliable synthetic transitions with a soft confidence rule. We evaluate UC-VT across three tiers of increasing realism: (i) a one-step contextual-bandit proof of concept (100 seeds), (ii) a multi-step, Fitted-Q-Iteration sequential proof of concept (25 seeds), and (iii) a continuous-control pilot on a SAC-based deep meta-RL agent (`cheetah_vel`, single seed). Across both toy tiers, UC-VT is substantially and significantly safer than unweighted, randomly weighted, or uniformly matched-mass virtual training. Against real-data-only training, UC-VT is significantly better in the one-step tier and statistically indistinguishable in the sequential tier. In the deep-RL pilot, UC-VT matches or modestly exceeds real-only and unweighted baselines over an early, matched training window, but extended training reveals a critic-loss divergence coincident with large spikes in the virtual Bellman loss – a pattern consistent with the failure mode anticipated by our theoretical bound when uncertainty becomes miscalibrated at extrapolated latents, and one our fixed-radius pilot implementation does not yet guard against. We report this honestly as a central finding: the toy-scale mechanism is validated, while the deep-RL pilot highlights the importance of evaluating the uncertainty-controlled curriculum from our original proposal as a stability mechanism at scale.

## 1 Introduction

Meta-reinforcement learning (meta-RL) aims to train an agent that can adapt rapidly to new tasks drawn from a distribution, but most methods train and test on the same distribution, leaving performance under distribution shift largely untested. Latent Dynamics Mixture (LDM) [1] broadens the effective training support by generating imaginary tasks, and Task-Aware Virtual Training (TAVT) [2] improves the quality of those virtual tasks with task-aware latent geometry, task-preserving generation, and state regularization. Both leave a shared vulnerability: a generated task can preserve a plausible latent identity while its decoded reward or transition is simply wrong. Training uniformly on such a sample biases the Bellman target; rejecting it outright is safer but discards moderately novel tasks that could otherwise be useful for extrapolation.

**Research question.** Can calibrated decoder uncertainty control synthetic-task influence so that out-of-distribution (OOD) coverage increases without accepting unreliable virtual transitions?

**Contributions.** (1) We formalize UC-VT: an ensemble task decoder, a soft uncertainty-based confidence weight on the virtual loss, and an uncertainty-controlled curriculum for the virtual-task radius, together with a bound on the resulting virtual-gradient bias. (2) We validate the core mechanism in a one-step contextual bandit (100 seeds) and a multi-step sequential decision task solved by Fitted-Q Iteration (25 seeds), in both cases with matched-mass, random-weight, and hard-rejection controls and an unattainable oracle diagnostic. (3) We report a continuous-control pilot built on a from-scratch SAC + TAVT-style meta-RL agent, and we honestly report a critic-divergence failure mode that appears only at this scale, showing that it is consistent with a breakdown of the same calibration assumption our theoretical bound depends on. We view (3) as the paper's most useful empirical contribution: it shows precisely where a toy-validated mechanism needs more machinery before it can be trusted in deep RL.

## 2 Related Work

LDM [1] mixes imagined tasks from a latent dynamics model into meta-training. TAVT [2] is our direct base learner: it adds a task-preserving (bisimulation-style) loss that regresses inter-task latent distance against inter-task reward/transition distance, and a state regularization term intended to keep virtual states on a realistic manifold. Deep ensembles [3] are a standard, inexpensive way to obtain calibrated epistemic uncertainty without a fully Bayesian model, and are the basis of our decoder ensemble.

## 3 Method

### 3.1 Ensemble task decoder and uncertainty score

We replace TAVT's single task decoder with $K$ independently initialized decoders,

$$D_k(s, a, z) = (\hat{r}_k, \Delta\hat{s}_k), \qquad k = 1, \ldots, K,$$

which map a state, action, and task latent $z$ to a predicted reward and next-state delta. The ensemble mean is used to generate the virtual transition; ensemble disagreement gives an epistemic uncertainty score,

$$u = \mathrm{Var}_k(\hat{r}_k) \; + \; \lambda_s \frac{1}{d_s} \sum_j \mathrm{Var}_k(\Delta\hat{s}_{k,j}),$$

combining reward-prediction variance with a scaled, dimension-averaged next-state-prediction variance ($d_s$ is the state dimension).

### 3.2 Confidence weighting

The default soft rule down-weights the virtual loss in proportion to uncertainty, relative to a running scale $c_t$ (detached from the computation graph):

$$w(u) = \exp(-\beta u / c_t), \qquad \mathcal{L} = \mathcal{L}_{\text{real}} + \lambda_v \, \mathbb{E}[w(u)\, \mathcal{L}_{\text{virtual}}] \, .$$

$c_t$ is a running percentile (65th, by default) of recently observed uncertainty values. A thresholded gate $w(u) = \mathbb{1}[u \le \tau]$ is retained as the "hard rejection" ablation, since the best bias-coverage trade-off may depend on horizon and shift type.

### 3.3 Uncertainty-controlled curriculum

Candidate task latents are generated by interpolation and controlled extrapolation between encoded training-task latents. The proposal is that the maximum extrapolation radius $\rho_t$ should increase only when held-out decoder error and ensemble disagreement remain positively calibrated – i.e., only when higher predicted uncertainty reliably tracks higher true error. **We flag this explicitly: our deep-RL pilot implementation generates virtual latents from a fixed interpolation/extrapolation range and does not implement this calibration-gated radius schedule.** As Section 4.3 shows, this omission is very likely responsible for the instability we observe once training runs long enough for the decoder ensemble to be queried on regions where its uncertainty estimate is no longer trustworthy.

### 3.4 Theoretical justification

Let $y(x)$ be the ideal Bellman target for input $x$, $\hat{y}(x)$ its ensemble-mean estimate, and $u(x)$ the disagreement above. If the estimator is calibrated, $|\hat{y}(x) - y(x)| \leq C\sqrt{u(x)}$, and the critic-loss gradient is $L$-Lipschitz in its target, then for any confidence rule $0 \leq w(u) \leq 1$,

$$\left\| \nabla \mathcal{L}_v^{\hat{w}} - \nabla \mathcal{L}_v^{w} \right\| \;\leq\; LC\,\mathbb{E}\big[w(u)\sqrt{u}\big] \;\leq\; LC\,\mathbb{E}\big[\sqrt{u}\big]\,.$$

Since $w \in [0, 1]$, confidence weighting cannot increase this upper bound relative to unweighted training, and under a $\gamma$-contractive Bellman operator the resulting value-error perturbation scales with $1/(1-\gamma)$. This is a safety argument, not a convergence guarantee, and it is conditional on the calibration assumption $|\hat{y} - y| \leq C\sqrt{u}$ holding at the inputs actually queried. Section 4.3 presents evidence consistent with this assumption failing at scale when the curriculum in Section 3.3 is not enforced.

## 4 Experiments

We evaluate UC-VT across three tiers of increasing realism, using the same comparison set throughout wherever the tier makes it meaningful: real-only training, unweighted virtual training, a uniform matched-mass control (same total synthetic weight as UC-VT, distributed evenly), randomly permuted uncertainty weights, hard rejection, UC-VT soft weighting, and (where an oracle is computable) an unattainable diagnostic that allocates the same synthetic weight budget to the lowest true-error samples.

### 4.1 Tier 1: One-step contextual bandit (100 seeds)

Task $z$ has reward $r_z(a) = -\big(a - [\sin(1.3z) + 0.15z]\big)^2 + \epsilon$. Training tasks use $z \in [-0.6, 0.6]$; OOD tests span $|z| \in [0.7, 1.5]$ across near/mid/far bins. The best action per training task is estimated from noisy interaction (a local quadratic fit), never from the simulator's true optimum, which is used only for evaluation and the oracle diagnostic. Nine bootstrap polynomial reward models generate virtual pseudo-actions; action variance is the uncertainty score.

Per-seed uncertainty/true-error correlation was $0.774 \pm 0.166$ (Pearson) and $0.867 \pm 0.123$ (Spearman) – the uncertainty signal is a genuinely informative proxy for decoder error in this setting. UC-VT improved over unweighted VT by 0.355 (paired 95% CI $[0.281, 0.429]$, $p < 10^{-14}$, won 95% of seeds) and over real-only by 0.070 (CI $[0.041, 0.099]$, won 90%). Hard rejection was 0.013 better than soft UC-VT ($p = 0.0037$) – an informative limitation, not a hidden negative result:

Table 1: Tier 1 (bandit): OOD return, mean ± std over 100 seeds.

| Method | OOD return ↑ |
|---|---|
| Real only | $-0.149 \pm 0.032$ |
| Unweighted VT | $-0.433 \pm 0.418$ |
| Random weights | $-0.440 \pm 0.442$ |
| Hard rejection | $-0.065 \pm 0.141$ |
| **UC-VT** | $\mathbf{-0.079 \pm 0.147}$ |
| Oracle diagnostic | $-0.023 \pm 0.033$ |

this one-step task has no long-horizon exploration cost, so aggressive pruning can be sufficient here specifically.

### 4.2 Tier 2: Multi-step sequential proof of concept (25 seeds)

To test whether the mechanism survives temporal credit assignment, we extend the bandit to a short-horizon ($H = 6$, $\gamma = 0.9$) goal-reaching task: the agent controls a 1-D state via $s' = \text{clip}(s + 0.5a)$ (known, deterministic dynamics), and per-step reward $-(s - g(z))^2 - 0.05a^2$ depends on a hidden, task-specific goal $g(z) = \sin(1.3z) + 0.15z$. Because large single-step corrections are penalized, the optimal policy must plan several steps ahead rather than jump directly to the goal, giving genuine temporal structure absent from Tier 1. We use Fitted-Q Iteration (ridge regression on polynomial features of $(s, a, z)$, closed-form, no neural networks) so the tier isolates the sequential-credit-assignment question without confounding it with deep-network optimization difficulties – that question is addressed separately in Tier 3.

Table 2: Tier 2 (multi-step): discounted OOD return, mean ± std over 25 seeds.

| Method | OOD return ↑ |
|---|---|
| Real only | $-2.939 \pm 0.413$ |
| Unweighted VT | $-4.386 \pm 0.040$ |
| Uniform matched-mass | $-4.396 \pm 0.085$ |
| Random weights | $-4.411 \pm 0.176$ |
| Hard rejection | $-4.112 \pm 0.184$ |
| **UC-VT** | $\mathbf{-3.069 \pm 0.338}$ |
| Oracle diagnostic | $-4.045 \pm 0.075$ |

Per-seed correlation between uncertainty and true decoder error remained strong ($0.802 \pm 0.132$ Pearson, $0.885 \pm 0.027$ Spearman), essentially unchanged from Tier 1. UC-VT dominates every naive virtual-training baseline by a wide, highly significant margin: $+1.317$ over unweighted VT (95% CI $[1.190, 1.443]$, $p = 1.1 \times 10^{-16}$, won 100% of seeds), and similarly large, significant wins against uniform matched-mass, random weighting, and hard rejection (all $p < 10^{-12}$, 100% win rate). Against real-only, however, UC-VT is statistically indistinguishable: mean difference $-0.130$ (95% CI $[-0.306, 0.045]$, $p = 0.159$), with real-only winning on 64% of seeds. Across both toy tiers, UC-VT reliably rescues virtual training from being actively harmful; relative to real-only training, it is significantly better in Tier 1 and statistically indistinguishable in Tier 2.

### 4.3 Tier 3: Continuous-control pilot (SAC + TAVT-style meta-RL, single seed)

**Setup.** We implement a from-scratch Soft Actor-Critic (SAC) agent with a permutation-invariant context task encoder, twin critics with target networks, a bisimulation-style task-preserving loss, a Lipschitz-style state-regularization penalty, and the UC-VT ensemble decoder and soft weighting described above, trained on `cheetah_vel` – a HalfCheetah task family in which the reward penalizes deviation from a task-specific target forward velocity, matching the reward-shift character of TAVT's original `cheetah-vel-ood` benchmark. We compare three configurations that differ only in two hyperparameters: **UC-VT** (full method), **Real-only** ($\lambda_v = 0$, no virtual transitions), and **Unweighted** ($\beta = 0$, so $w(u) \equiv 1$ and all virtual transitions are trusted equally). *This pilot uses a single random seed per configuration and should be read as an initial feasibility study, not a benchmark-grade result* – Tiers 1-2 are where our statistically powered claims live.

**An honest confound.** The UC-VT training log continues to iteration ∼3990 (with evaluations through iteration 3950), whereas the Real-only and Unweighted evaluation logs end at iteration 1950. We therefore report two things separately rather than mixing them: a fair, matched-window comparison over the range all three evaluation logs share (iterations 1500-1950), and a separate account of what happened in UC-VT's additional, unmatched training.

Table 3: Tier 3 (`cheetah_vel`): discounted return by evaluation bin, matched window (iterations 1500-1950), single seed. Mean OOD averages the near/mid/far columns.

| Method | Train | Near OOD | Mid OOD | Far OOD | Mean OOD ↑ |
|---|---|---|---|---|---|
| Real-only | −46.71 | −112.64 | −174.85 | −241.43 | −176.31 |
| Unweighted | −58.51 | −125.61 | −177.79 | −241.76 | −181.72 |
| **UC-VT** | −50.97 | **−106.14** | **−171.99** | **−236.30** | **−171.48** |

Within this matched window, UC-VT modestly outperforms both baselines on every OOD bin, consistent in direction (though obviously not in statistical power, at $n = 1$ seed) with Tiers 1-2. This is the result we would have reported as the paper's headline finding if we stopped measuring here.

**What happens with more training.** Table 4 and Figure-equivalent description below show UC-VT's own late-training window (iterations 3500-3950, no baseline comparison possible since those runs were not extended this far).

Table 4: UC-VT only: early vs. late training window, mean discounted return.

| Window | Train | Mean OOD | Mean real critic loss / mean virtual critic loss |
|---|---|---|---|
| Early (it. 1500-1950) | −50.97 | −171.48 | 0.594 / 0.000605 |
| Late (it. 3500-3950) | −79.43 | −216.56 | 251.07 / 3426.64 (max 35,246) |

Mean OOD return degrades by roughly 45 points between the two windows, and the logged real critic loss rises by more than two orders of magnitude, with the (already confidence-weighted) virtual critic loss occasionally spiking into the tens of thousands within a single logging interval. Inspecting the raw training log (iterations 3900-3990) shows this is not a smooth drift but a series of sharp spikes – virtual critic loss varies from about 26 to more than $3.5 \times 10^4$ across consecutive

10-iteration logging windows, with abrupt multi-thousand spikes and drops, while the real critic loss also becomes highly variable. The pattern is consistent with individual virtual transitions occasionally carrying a badly wrong decoded reward or next state that the soft weight $w(u)$ failed to sufficiently suppress, corrupting the shared critic for both real and virtual updates on that step.

# 5 Discussion

## 5.1 Why does virtual training with weighting help in the toy tiers?

In both the bandit and the multi-step tier, the uncertainty signal is strongly and consistently correlated with true decoder error (Pearson $> 0.77$ in both tiers). This is precisely the condition the bound in Section 3.4 requires: when $u(x)$ reliably tracks $|\hat{y}(x) - y(x)|$, down-weighting high-$u$ samples provably cannot increase gradient bias relative to using them unweighted, and empirically it recovers most of the benefit of hard rejection without discarding as much data – which is exactly why UC-VT dominates unweighted, random, and uniform-matched training by a wide, significant margin in both toy tiers, while outperforming real-only in Tier 1 and closing the gap to statistical indistinguishability in Tier 2.

## 5.2 Why does the same rule destabilize at deep-RL scale?

The theoretical bound is conditional on calibration holding *at the inputs the policy actually visits*. In the toy tiers, the virtual-task latent space is low-dimensional and densely covered by the training-task range, so the ensemble's disagreement is a good proxy for its error almost everywhere it is queried. In the deep-RL pilot, the policy is simultaneously improving and exploring new regions of a much higher- dimensional state space, and virtual latents are drawn from a *fixed*, non-curriculum interpolation/extrapolation range (Section 3.3) rather than one whose radius is gated on held-out calibration, as our own proposal specifies. Once the decoder ensemble is queried far enough outside the region where its disagreement was calibrated during training, $|\hat{y} - y| \leq C\sqrt{u}$ can fail – the ensemble can be confidently wrong, producing a low $u$ (and therefore a large $w(u)$) for a transition whose target is nonetheless badly off. A small number of such transitions is enough to corrupt a shared critic network, which is consistent with the sharp, intermittent loss spikes we observe rather than a smooth degradation.

## 5.3 Sample efficiency and convergence

Over the matched early window, all three configurations show comparable convergence behavior (critic loss $< 1$, stable actor loss, alpha annealing smoothly), and the training curves do not show virtual data providing a sample-efficiency advantage in the amount of environment interaction used in this pilot – consistent with the toy tiers' finding that the largest, most reliable benefit of UC-VT is *safety relative to naive virtual training*, not necessarily an outright improvement on strong real-data baselines with this amount of data. This should be re-examined with more environment steps and multiple seeds before drawing a stronger conclusion.

# 6 Limitations

- **Single seed in Tier 3.** All Tier-3 numbers come from one training run per configuration; we cannot report confidence intervals or significance for the deep-RL comparison the way we

can for Tiers 1-2. The matched-window result in Table 3 should be read as directional, not conclusive.

- **Mismatched training length.** UC-VT ran roughly twice as long as the two baselines. We restricted our primary comparison to the shared iteration range specifically to avoid this confound, but we have not verified whether Real-only or Unweighted would also destabilize given the same additional training budget.
- **No calibration-gated curriculum in the Tier-3 implementation.** Section 3.3 of our proposed method specifies that the virtual-task radius should expand only when calibration holds; our pilot implementation samples from a fixed radius. The observed divergence is consistent with this missing calibration safeguard, so implementing the gated curriculum is our highest-priority next step.
- **Simplified TAVT components.** Our task-preserving loss omits the official TAVT implementation's WGAN discriminator/generator refinement step and auxiliary context-distribution VAE; state regularization is a Lipschitz-smoothness penalty rather than the official mechanism. These are engineering simplifications made under a tight timeline, not part of the UC-VT contribution itself, but they mean Tier 3 is not a literal reproduction of TAVT's published training procedure.
- **Single task family.** We evaluate the reward-shift family (`cheetah_vel`) but do not report a Tier-3 transition-shift family (`walker_mass`) result in this draft; our codebase supports it but we prioritized finishing and stabilizing one full pipeline given the project timeline.

# 7 Conclusion and Future Work

We set out to test whether calibrated decoder uncertainty can make virtual training in meta-RL safer without sacrificing the OOD coverage virtual tasks are meant to provide. Across two well-powered toy tiers, the answer is a clear yes: UC-VT is dramatically and significantly safer than uniform, random, or matched-mass virtual training, and statistically competitive with using real data alone. Our deep-RL pilot extends this picture in a genuinely useful way rather than a purely positive one: over a training budget matched to two baselines, UC-VT shows the same modest, direction-consistent advantage seen at toy scale; but with additional training, we observe a critic-loss divergence consistent with a breakdown of the calibration assumption our theoretical bound depends on, together with the absence of the calibration-gated curriculum in our pilot implementation. We view identifying this failure mode – rather than papering over it – as the paper's most concrete contribution toward making UC-VT usable at scale.

Future work, in priority order: (1) implement the calibration-gated virtual- task radius from Section 3.3 and re-run the Tier-3 comparison to test whether it prevents the late-training divergence; (2) run Tier 3 with multiple seeds and a matched training budget across all three configurations to obtain a statistically powered deep-RL comparison; (3) extend Tier 3 to the transition-shift family (`walker_mass`); (4) reintroduce the omitted TAVT components (WGAN state regularization, context-distribution VAE) to test whether they independently affect stability; and (5) report the additional metrics specified in our original proposal – CVaR@20%, critic-target bias, and uncertainty-calibration curves – which we did not have time to compute for this draft but which would directly test the calibration-failure hypothesis in Section 4.3.

## References


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